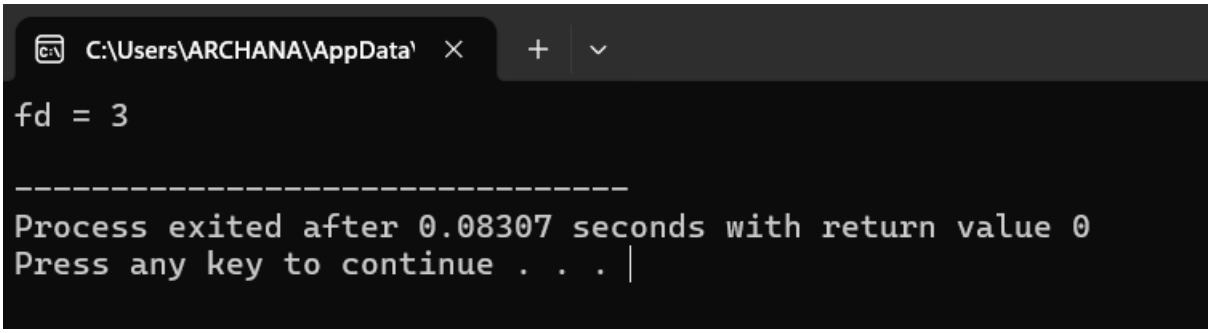
Question-25:

Construct a C program to implement the I/O system calls of UNIX (fcntl, seek, stat, opendir, readdir) .

PROGRAM:

```
#include<stdio.h>
#include<fcntl.h>
#include<errno.h>
#include<unistd.h>
extern int errno;
int main()
{
int fd=open("foo.txt",O_RDONLY|O_CREAT,0644);
printf("fd = %d\n",fd);
if(fd==-1)
{
printf("Error Number %d\n",errno);
perror("Program");
}
else
{
close(fd);
}
return 0;
}:
```

OUTPUT:



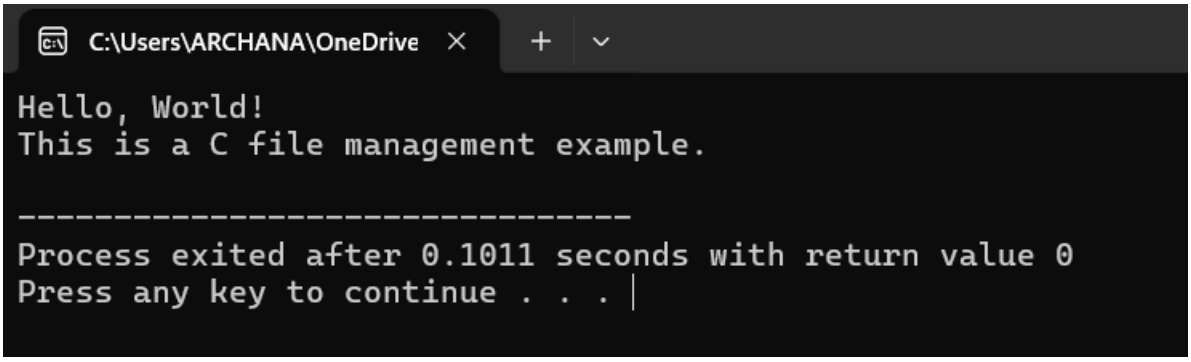
Question-26:

Construct a C program to implement the file management operations.

PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    FILE *file;
    file = fopen("example.txt", "w");
    if (file == NULL) {
        printf("Error opening the file for writing.\n");
        return 1;
    }
    fprintf(file, "Hello, World!\n");
    fprintf(file, "This is a C file management example.\n");
    fclose(file);
    file = fopen("example.txt", "r");
    if (file == NULL) {
        printf("Error opening the file for reading.\n");
        return 1;
    }
    char buffer[100];
    while (fgets(buffer, sizeof(buffer), file) != NULL) {
        printf("%s", buffer);
    }
    fclose(file);
    return 0;
}
```

OUTPUT:



Question-27:

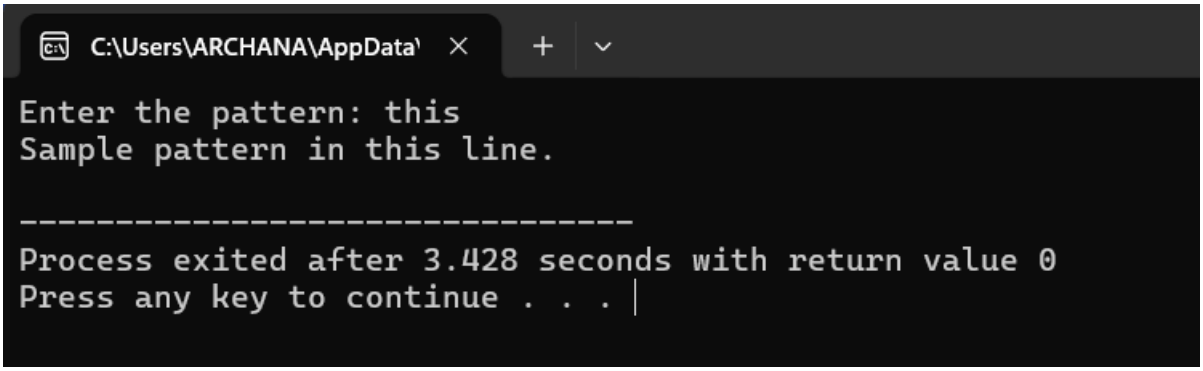
Develop a C program for simulating the function of ls UNIX Command.

PROGRAM:

```
#include<stdio.h>
#include<string.h>
int main()
{
char fn[200],pat[100],temp[200];
FILE *fp;
strcpy(fn,"C:\\Users\\ARCHANA\\OneDrive\\Pictures\\Documents\\Desktop\\OS\\sample.txt
");
printf("Enter the pattern: ");
scanf("%99s",pat);
fp=fopen(fn,"r");
if(fp==NULL)
{
perror("File opening failed");
return 1;
}
while(fgets(temp,sizeof(temp),fp)!=NULL)
{
if(strstr(temp,pat)!=NULL)
```

```
printf("%s",temp);
}
fclose(fp);
return 0;
}
```

OUTPUT:



Question-28:

Write a C program for simulation of GREP UNIX command.

PROGRAM :

```
#include<stdio.h>
#include<string.h>
#define MAX_LINE_LENGTH 1024
int main()
{
char pattern[100],line[MAX_LINE_LENGTH];
FILE *file;
file=fopen("C:\\Users\\ARCHANA\\OneDrive\\Pictures\\Documents\\Desktop\\OS\\sample.t
xt","r");
if(file==NULL)
{
perror("Error opening file");
return 1;
```

```
}
printf("Enter the pattern: ");
scanf("%99s",pattern);
while(fgets(line,sizeof(line),file)!=NULL)
{
if(strstr(line,pattern)!=NULL)
printf("%s",line);
}
fclose(file);
return 0;
}
```

OUTPUT:

